DGMS 2019 • SECTION E • QUESTION 29

Deltaic Environments: Subaqueous Delta Zonation, Delta Front & Prodelta Architecture

Directorate General of Mines Safety (DGMS) Junior Scientific Officer Examination 2019 | Sedimentology & Deltaic Depositional Systems Guide

Question Statement

Q.29. The uppermost part of subaqueous delta that lies at water depth down to 10 m or so, is referred to as:

a (a) Upper delta plain

b (b) Lower delta plain

c (c) Prodelta

d **(d) Delta front**

Correct Answer: (d) Delta front (Uppermost subaqueous delta, depth 0 to ~10 m)

Official Textbook Figure (Sam Boggs Jr., Principles of Sedimentology & Stratigraphy)

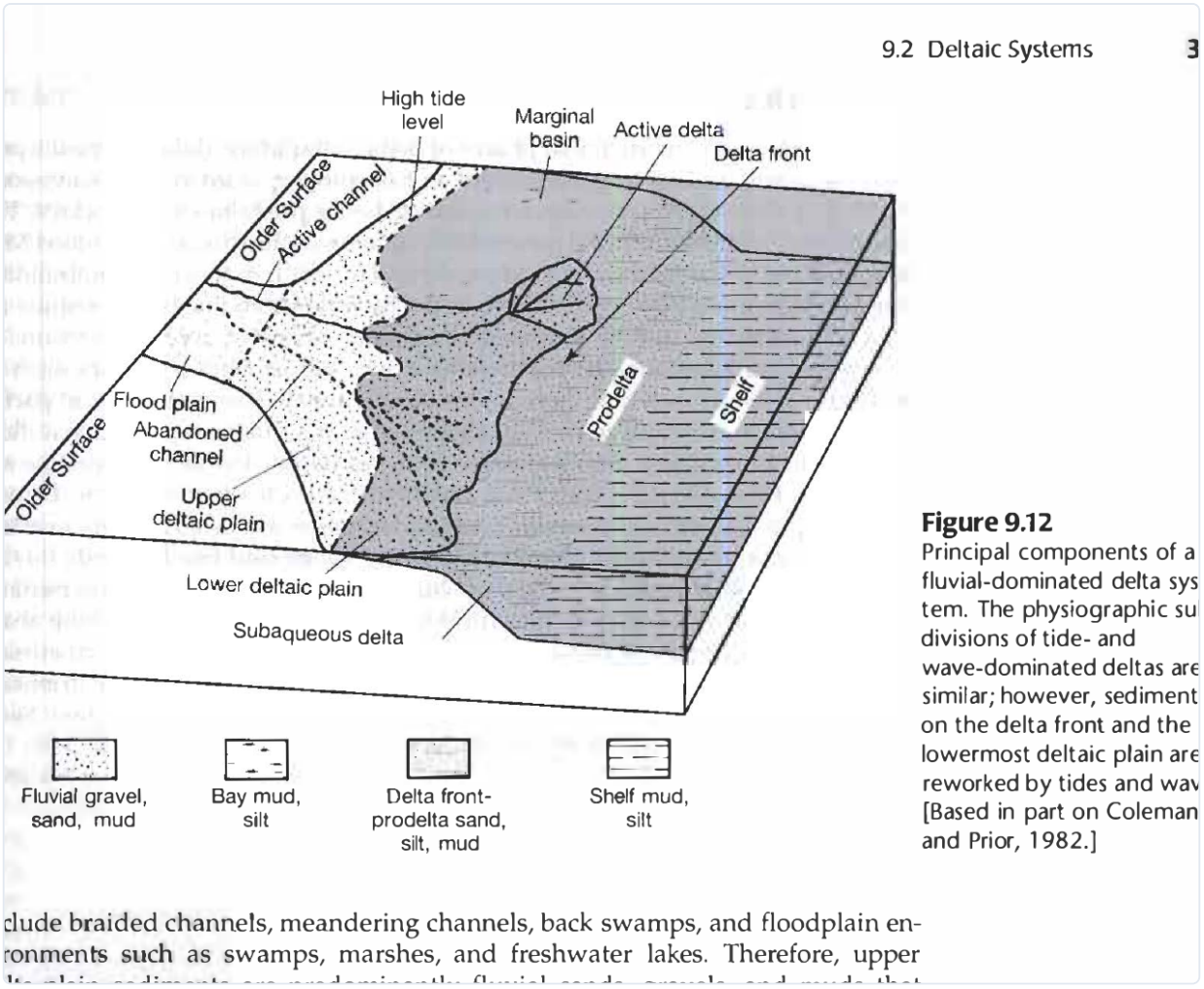


Figure 1: Sam Boggs Jr., Chapter 9, Figure 9.12. Principal physiographic components and facies of a delta system. Note the major boundary between the **Subaerial Delta Plain** (Upper Delta Plain above high-tide level, Lower Delta Plain in the intertidal zone) and the **Subaqueous Delta**, which is subdivided into the **Delta Front** (uppermost part down to ~10 m depth; sand and silt mouth bars) and the **Prodelta** (deeper seaward slope down to 200–300 m; hemipelagic mud and silt).

In-Depth Option Evaluation & Petrographic Analysis

(d) Delta front — [CORRECT]

Verbatim Definition from Sam Boggs Jr. (p. 316):

"The subaqueous delta plain lies seaward of the lower deltaic plain below low-tide

water level. ***The uppermost part of the subaqueous delta, lying at water depths down to 10 m or so, is commonly called the delta front."***

Hydrodynamics & Sediments: The delta front is the high-energy seaward interface where river effluents meet standing basin water. Sediments consist dominantly of **distributary-mouth-bar sands, sheet sands, and coarse silts** with cross-bedding and ripple laminations, winnowed by waves and tidal currents.

(c) Prodelta — [INCORRECT]

Characteristics: The prodelta is the *deeper seaward part* of the subaqueous delta lying beyond the delta front. It extends from ~10–20 m down to water depths of **200–300 m** where it merges into the open marine shelf.

Sediments: Lacks coarse sands; dominated by hemipelagic suspension deposition of fine clays, muds, and thin silt turbidites.

(a) Upper delta plain — [INCORRECT • SUBAERIAL]

Characteristics: Part of the **subaerial delta**, not the subaqueous delta. It lies completely above high-tide water level and is governed strictly by subaerial fluvial processes (braided/meandering channels, point bars, natural levees, backswamps, and freshwater marshes).

(b) Lower delta plain — [INCORRECT • INTERTIDAL SUBAERIAL]

Characteristics: Extends between the low-tide mark and the upper limit of tidal influence. While periodically inundated during high tide, it is subaerially exposed at low tide and features interdistributary bays, mangrove swamps, salt marshes, and tidal channels. It is not part of the continuously submerged subaqueous delta.

Structural Architecture & Facies of Classic Delta Systems

Realm	Subdivision	Depth / Elevation	Dominant Processes	Diagnostic Facies / Sediments
Subaerial Delta (Deltaic Plain)	**Upper Delta Plain**	Above high-tide level (Landward)	Fluvial channels, overbank floods, crevasse splay	Fluvial sands, gravels, floodplain muds, coal/peat
	Lower Delta Plain	Intertidal zone (Between low & high tide)	Mixed fluvial and tidal currents, marine inundation	Interdistributary bay muds, tidal creek sands, mangrove peat
Subaqueous Delta (Below Low Tide)	**Delta Front**	**Below low tide down to ~10 m depth**	**River-jet deceleration, wave winnowing, longshore drift**	**Distributary-mouth-bar sands, sheet sands, cross-bedded sandstones**
	Prodelta	From ~10 m down to 200–300 m (Seaward slope)	Low-energy suspension fallout, gravity slumps, turbidity currents	Dark marine muds, clays, bioturbated silts, prodelta claystones

Direct Textbook Quotations & Citations

Sam Boggs Jr. (2006), *Principles of Sedimentology and Stratigraphy, 4th ed., Chapter 9, p. 316:*

"The subaqueous delta plain lies seaward of the lower deltaic plain below low-tide water level. **The uppermost part of the subaqueous delta, lying at water depths down to 10 m or so, is commonly called the delta front.** *The remaining seaward part of the subaqueous delta is called the* **prodelta***, or prodelta slope... The prodelta may extend to water depths as much as 200–300 m."*

— Sam Boggs Jr., Chapter 9, p. 316

J. M. Coleman & D. B. Prior (1982), *Deltaic Sand Bodies, AAPG Continuing Education Course Note Series 15:*

"The delta front environment encompasses the subtidal delta platform from the low water datum to water depths of 10–12 meters. It represents the zone of most active sand deposition, forming distributary-mouth-bar complexes."

— Coleman & Prior (1982)

Exam Traps & Rapid Elimination Strategy

Rule of Thumb for Deltaic Questions in UPSC & DGMS:

1. **Subaerial vs. Subaqueous:** If the question mentions *subaqueous*, immediately eliminate **Upper Delta Plain** and **Lower Delta Plain**.
2. **Delta Front vs. Prodelta:**
   - **Delta Front:** Shallow (0 to ~10 m depth), high energy, sandy, mouth bars.
   - **Prodelta:** Deep (10 m to 300 m depth), low energy, muddy/clayey, prodelta slope.